

FPGA Session Log — **final_8**: 4-to-16 Decoder

Digilent Arty A7-35T · Open-Source Toolchain

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July 13, 2026

1 Session Goal

Compile the hand-written `top.v` (a 4-to-16 decoder mapping two 2-bit inputs $\{A2, A1\}$ and $\{B2, B1\}$ to a seven-segment pattern `seg[6:0]`), create a testbench, and establish the simulation flow. The Pmod SSD (seven-segment display) hardware is on order.

2 Compilation Result

Synthesis was attempted with Yosys 0.66 (oss-cad-suite):

```
yosys -p "synth_xilinx -flatten -abc9 -arch xc7 -top top; \  
        write_json top.json" top.v
```

It failed at the first parse error:

```
top.v:2: ERROR: syntax error, unexpected TOK_ID,  
        expecting ')' or ',' or '='
```

Three recurring syntax issues were identified (fixes to be applied by the author):

1. **Port list commas** (line 2): ports must be comma-separated, including between the `input` and `output` declarations, e.g. `input wire A2, A1, B2, B1,.`
2. **Case-item colons** (lines 9–25): each case item requires a colon, e.g. `4'b0000: seg = ...` (likewise `default:`).
3. **Statement semicolons**: every assignment `seg = 7'b...` must end with `;`.

The decoder logic itself (the 2-bit $\times$ 2-bit product table) is expected to compile cleanly once these are corrected.

3 Testbench

`tb_top.v` was created. It sweeps all 16 combinations of $\{A2, A1, B2, B1\}$, prints the resulting `seg[6:0]` pattern for each, and dumps `tb_top.vcd` for waveform viewing. It is deliberately *not* self-checking: the printed patterns are to be compared by hand against the case table in `top.v`.

4 Simulation Flow

From `projects/final_8/`:

```
source /Users/lucaszhou/15A/FPGAproj1/tools/oss-cad-suite/environment
iverilog -g2005 -o tb_top.vvp tb_top.v top.v # compile
vvp tb_top.vvp # run, prints 16 cases
gtkwave tb_top.vcd # optional waveforms
```

5 Follow-up Session (same day): XDC Rewrite

The author fixed the case-item colons and statement semicolons in `top.v`; one syntax issue remains (port-list commas, line 2). `arty.xdc` was then rewritten to match the current ports:

- **Inputs:** A2/A1/B2/B1 on the slide switches, keeping the earlier assignment A2=SW3 (A10), A1=SW2 (C10), B2=SW1 (C11), B1=SW0 (A8).
- **Outputs:** `seg[6:0]` on Pmod connector **JA** (top-right of the board), where the Pmod SSD's segment inputs land: `seg[0]`→JA1 (G13, signal AA) through `seg[6]`→JA9 (A18, signal AG); i.e. `seg[N]` drives SSD segment $N+1$. If a segment lights in the wrong position on real hardware, permute the XDC pin assignments rather than the `top.v` case table.
- **Digit select:** the Pmod SSD is a two-digit display; JA pin 10 (K16, signal C) selects the lit digit. The constraint is present but commented out because `top.v` has no such port yet.

Verification: a scratch copy of `top.v` (with only the line-2 commas added; the author's file was not modified) was run through Yosys and nextpnr-xilinx with the new XDC — clean synthesis, all 11 ports constrained to their pads, and a FASM file generated. The real build should therefore pass once the author applies the comma fix.

Adding the digit-select port (to be applied by the author)

```
module top (
  input wire A2, A1, B2, B1, //bits
  output reg [6:0] seg, //from 1 to 7
  output wire C //digit select for Pmod SSD
);
...
assign C = 1'b1; // outside the always block; flip to 1'b0 if the wrong digit lights
```

After adding the port, uncomment the K16 line at the bottom of `arty.xdc`.

6 Open Items

- **Fix `top.v` port commas** (author) — line 2: `input wire A2, A1, B2, B1,.`
- **Add digit-select port C** (author) — snippet above; then uncomment the K16 constraint in `arty.xdc`.
- **Confirm on hardware once the Pmod SSD arrives** — segment ordering (`seg[N]` → segment $N+1$ assumed) and digit-select polarity.

- **Fix `tools/env.sh` under `zsh`** — its `BASH_SOURCE` fallback points at a stale path from an old machine; as a workaround the `oss-cad-suite` environment was sourced directly.

7 Files in `projects/final_8/`

File	Status	Notes
<code>top.v</code>	one syntax fix left	written by author; line-2 commas pending
<code>tb_top.v</code>	new this session	16-case sweep + VCD dump
<code>arty.xdc</code>	rewritten	Pmod SSD on JA; verified through P&R
<code>build_and_flash.sh</code>	existing	end-to-end build script